

1. Initial Expectation of the Outcome

Given the design premise of FairLink—to identify cross-jurisdictional differences in algorithmic fairness compliance—my initial expectation was to observe differential compliance outcomes for the same dataset across Hong Kong and the U.S. Specifically, I anticipated a scenario where the disparity rate would exceed one jurisdiction’s threshold but remain below the other’s, thereby demonstrating how identical model behavior can trigger divergent regulatory assessments.

In this case, however, the observed 55% disparity gap far exceeded both the Hong Kong (18%) and U.S. (10%) thresholds, resulting in a “double non-compliance” flag for both jurisdictions. While this outcome did not align with my initial expectation of a divergent result, it provided a more robust illustration of the tool’s core functionality: its ability to simultaneously validate against multiple regulatory frameworks, regardless of whether the final outcomes converge or diverge.

2. Rationale for Not Adjusting Synthetic Data to Create a Divergent Scenario

- **Preservation of Methodological and Ethical Integrity**

The primary purpose of this prototype is to deliver objective, data-driven compliance insights. Adjusting the dataset solely to create a dramatic “cross-jurisdictional divergence” would undermine this principle by prioritizing a performative demonstration over realistic risk assessment. Such a modification would not reflect real-world business conditions, where disparities often exceed the strictest applicable threshold and trigger non-compliance across multiple jurisdictions. By retaining the unaltered 55% gap, I ensured the prototype’s output remained grounded in a realistic high-risk scenario, preserving its credibility as a tool for identifying genuine regulatory exposure.

- **Demonstration of the Tool’s Broader Utility Beyond “Divergence Detection”**

While cross-jurisdictional differences are a core selling point of FairLink, they are not its only value proposition. The “double non-compliance” outcome highlights the tool’s equally critical role in identifying convergent high-risk scenarios—where a single dataset fails to meet the requirements of all relevant jurisdictions. This is particularly relevant for a global institution like the Bank of East Asia, where a single lending model must satisfy both U.S.

fair lending laws and Hong Kong's anti-discrimination guidelines. The 55% gap case vividly illustrates how FairLink can serve as a unified "early warning system" for such universal compliance failures, addressing a far more urgent risk than isolated regulatory divergence.

- Avoidance of Misleading Stakeholder Perceptions

Manufacturing a divergent outcome could create a false impression that regulatory non-compliance is always a jurisdiction-specific problem, rather than a systemic risk rooted in poor model design. The unadjusted 55% gap, by contrast, delivers a clear, unambiguous message: when disparities are extreme, they will fail to meet even the most lenient regulatory standards. This is a far more impactful demonstration of the need for proactive fairness monitoring, and it directly aligns with the practical challenges the bank faces in ensuring its lending models meet the most stringent applicable rules across all markets.

In summary, while a divergent outcome would have showcased the tool's cross-jurisdictional differentiation capability, retaining the unaltered data provides a more comprehensive, credible, and practically relevant demonstration of FairLink's core purpose: to identify algorithmic fairness risks, whether they manifest as divergent or convergent compliance outcomes.